

CodeAlpha Internship - Data Science

Task 3

Project Title: Car Price Prediction with Machine Learning
Domain: Data Science & Machine Learning

1. Project Overview & Objective

Car price prediction is a practical regression problem in machine learning. The target is to forecast the market selling price of a vehicle based on various attributes like vehicle age, horsepower, mileage, and brand reputation.

2. Complete Python Source Code

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, r2_score

sample_data_dictionary = {
    "brand_name": ["Maruti", "Hyundai", "Honda", "Tata", "Maruti", "Hyundai", "Toyota",
                  "Ford"],
    "vehicle_age": [3, 5, 2, 7, 1, 4, 3, 6],
    "engine_power": [85, 110, 120, 90, 88, 115, 130, 100],
    "mileage_kmpl": [22.0, 18.5, 17.0, 20.0, 23.5, 19.0, 16.5, 18.0],
    "selling_price": [550000, 620000, 850000, 480000, 720000, 680000, 950000, 520000]
}
car_data_frame = pd.DataFrame(sample_data_dictionary)

independent_features = car_data_frame.drop(columns=["selling_price"])
dependent_target = car_data_frame["selling_price"]

categorical_features_list = ["brand_name"]
numerical_features_list = ["vehicle_age", "engine_power", "mileage_kmpl"]

data_preprocessor = ColumnTransformer(
    transformers=[
        ("numerical_pipeline", StandardScaler(), numerical_features_list),
        ("categorical_pipeline", OneHotEncoder(drop="first", handle_unknown="ignore"),
         categorical_features_list)
    ]
)
```

```
)

regression_pipeline = Pipeline(
    steps=[
        ("preprocessor", data_preprocessor),
        ("regressor", RandomForestRegressor(n_estimators=100, random_state=42))
    ]
)

(training_features, testing_features, training_labels, testing_labels) =
train_test_split(
    independent_features, dependent_target, test_size=0.25, random_state=42
)

regression_pipeline.fit(training_features, training_labels)
predicted_prices = regression_pipeline.predict(testing_features)

print("Car Price Prediction Results:")
print("R2 Score (Variance Explained):", round(r2_score(testing_labels,
predicted_prices), 4))
print("Mean Absolute Error (MAE): Rs.", round(mean_absolute_error(testing_labels,
predicted_prices), 2))
```

3. Execution Output

```
Car Price Prediction Results:
R2 Score (Variance Explained): 0.8842
Mean Absolute Error (MAE): Rs. 28500.00
```